



Industry effects of corporate environmental and social scandals: Evidence from China

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ABSTRACT

This study employs novel corporate environmental, social, and governance profiles to investigate the industry effects of environmental and social (ES) scandals in China. The findings reveal a notable decrease in stock prices for rival firms during scandal announcements. Further, we document a significant, positive correlation between rivals' ES performance and the abnormal returns over a five-day period surrounding the scandals. This correlation is more pronounced in rivals that disclose ES information. Additionally, relative to high-performing rivals, those with weaker ES performance significantly enhance their ES performance in the following year, driven by the perceived ES value in industry scandals. The findings also underscore the influence of state ownership, external governance environment, and industry competition on the spillover effects of ES scandals via risk channels.

1. Introduction

Concerns about Environmental, Social, and Governance (ESG) have grown over the years. When firms face environmental and social (ES) scandals, they often experience a double blow. These negative events can diminish firms' cash flows due to increased costs from equipment maintenance, adjustments, and fines, and decreased sales from reduced demand (Derrien et al., 2022). Moreover, equity investors, who have incorporated ESG into their investment strategies (Chava, 2014; Ng & Rezaee, 2015; Seltzer et al., 2022), may further penalise these firms by reducing their market value once the scandals become public (Kruger, 2015; Xu et al., 2012). Firms do not operate in a vacuum; they share and compete in product markets. Therefore, it is plausible that ES scandals could alter rivals' competitive positions in these markets and prompt investors to reassess ES risks across the industry. However, the externalities from ES scandals remain inadequately explored. Prior research has noted industry effects during corporate events such as bankruptcy (Lang & Stulz, 1992), takeovers (Shahrur, 2005), accounting restatements (Gleason et al., 2008), dividend announcements (Firth, 1996), and mergers and acquisitions (Fee & Thomas, 2004). This study

contributes to the literature by investigating how ES scandals impact the stock prices of industry competitors.

This research utilises a unique dataset of corporate ESG profiles from the International Institute of Green Finance (IIGF) in China. It integrates international standards with Chinese characteristics to form a '1 + 1' ESG rating system and includes negative behaviours and risk exposures to identify potential ESG risks. With the event study methodology, we document a significant average decline in the stock prices of rival firms for 615 ES scandals across 71 industries from 2014 to 2021. The mean shareholder wealth loss for these rivals was 0.43% within a window of two days before and after the scandal announcements. This finding suggests a dominant contagion effect over a competitive one, supporting the notion of an industry-wide risk effect rather than a firm-specific one arising from ES scandals. Our findings are robust, confirmed through rigorous endogeneity and robustness tests.

Additionally, we conduct a series of regression analyses to explore the common risk channel driving the contagion effect. Specifically, we assess how competitors' ES performance influences contagion in stock returns. Empirical evidence reveals a significant positive correlation between rivals' ES performance and their abnormal returns over five

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days surrounding scandal announcements. This correlation is more pronounced in rivals that disclose ES information. Moreover, rivals with weak ES performance significantly enhance their ES performance in the subsequent year relative to those with strong performance, driven by the value of ES activities during industry scandals. This finding underscores a potential incentive for firms to improve their ES performance.

To further substantiate our common risk argument, we conduct additional tests. Specifically, we hypothesise that if investors perceive industry rivals to share similar ESG risks with the event firms, the contagion effect would be more pronounced for rivals with high ESG risks. First, we consider the role of state ownership. Our empirical findings confirm that state ownership among rivals can mitigate the contagion effect of ES scandals, provided the event firms are not state-owned. Second, we examine the influence of external governance environments, which vary significantly across different regions of China. Our analysis, using the regional legal environment and economic development as proxies for external environments, reveals a stronger contagion effect of ES scandals when both event firms and their rivals operate in regions with weak external governance. Third, we investigate the impact of industry competition on returns surrounding scandals. The empirical evidence indicates that ES scandals adversely affect rivals in low-competition industries more than those in high-competition industries, thereby supporting our common risk hypothesis.

We focus on China, motivated by three primary reasons. First, the Chinese stock market is characterized by information asymmetry (Morck et al., 2000), leading to industry effects, particularly contagion effects, as external investors often depend on scandal-related information to assess other firms within the same industry. Second, China is transitioning towards sustainable development, with an increasing societal and governmental emphasis on ESG issues. Third, ESG practices in China are nascent, and ESG performance varies significantly among listed firms. This variation allows us to investigate how competitors' ES performance influences the spillover effect of ES scandals. Therefore, the selection of Chinese-listed firms as research subjects is apt. Moreover, China has forged trade relations globally and has been the world's largest exporter since 2014. This cooperation benefits both Chinese and international firms, fostering closer relationships. Additionally, the Chinese government is collaborating with the United Nations to expedite the sustainable development of the global economy. Thus, empirical evidence from the Chinese stock market could have substantial implications for managers, investors, and policymakers worldwide, particularly in countries with significant imports from China.

This study enriches the literature on industry effects. Lang and Stulz (1992) initially explore the industry impact of bankruptcy announcements. Yu et al. (2015) further investigate corporate scandals in China, identifying a contagion effect on industry rivals and confirming that good corporate governance can mitigate this negative impact. Kashmiri et al. (2017) find that customer data breach scandals adversely affect industry rivals, but this influence can be moderated by good CSR performance. Cao et al. (2019) analyse how rivals respond to firms' adoption of CSR proposals, revealing that rivals of firms narrowly passing CSR proposals experience lower announcement returns and higher subsequent-year CSR scores than those narrowly rejecting proposals. However, it remains unclear whether firms' ES scandals could impact industry rivals. This study addresses this gap by examining the industry effect of ES scandals.

This research also extends the literature on market reactions to negative ESG events. Xu et al. (2012) study environmental violation events among Chinese-listed firms, noting a small but negative impact on stock prices for event firms. Becchetti et al. (2012) demonstrate a significant negative effect on stock returns when firms exit the Domini 400 Social Index, a recognised CSR benchmark. Kruger (2015) analyses stock market reactions to ES news, documenting a negative investor response to adverse news. This study builds on this research by investigating the influence of ES scandals on industry rivals, not just event firms.

Additionally, this study contributes to the literature examining whether ESG enhances shareholder value during negative external shocks. Lins et al. (2017) show that firms with strong ESG performance experienced significantly higher stock returns during the Great Recession, attributed to the trust between firms and their investors and stakeholders. Recent literature examines the value of ESG during the COVID-19 pandemic, noting a positive relationship between ESG and stock performance (Albuquerque et al., 2020; Broadstock et al., 2021; Ding et al., 2021; Shan & Tang, 2023). This study broadens the scope of this literature by demonstrating that ES activities can also help mitigate the negative influence derived from industry ES scandals.

The remainder of this paper is structured as follows. Section 2 develops the hypotheses. Section 3 outlines our sample and methodology. Section 4 presents the empirical results of the industry effect of ES scandals. Section 5 conducts mechanism analyses. Section 6 examines industry rivals' responses to ES scandals. Finally, Section 7 concludes the study.

2. Hypothesis development

ES scandals can affect industry rivals through two primary mechanisms: contagion and competitive effects. Firms within the same industry often exhibit similar behaviours due to shared production processes and markets (Yu et al., 2015). For instance, following the exposure of melamine contamination in Sanlu Group's milk, over 20 domestic dairy plants were found to have excessive melamine levels. Such scandals may signal to investors the potential for similar risks across the industry, leading to a contagion effect that decreases the market value of industry rivals. Conversely, ES scandals could benefit industry rivals by shifting demand away from the scandal-hit firms. Specifically, a decline in corporate reputation and customer loyalty could render the products of the implicated firms unattractive, leading to a demand shift towards their industry rivals. This positive information could result in a competitive effect.

The impact of an ES scandal on its industry rivals is the sum of these two effects, with neither necessarily dominating (Lang & Stulz, 1992). If an ES scandal primarily results from an industry-common risk, the contagion effect will prevail. Conversely, the competitive effect will dominate if the scandal arises from a firm-specific risk. In general, the contagion effect is expected to be negative, and the competition effect positive. The overall influence of ES scandals on industry rivals, being a combination of these two effects, is an empirical question. We propose the following hypothesis to examine the actual effects of ES scandals on the stock returns of industry rivals:

H1. *ES scandal announcements will induce a decline in the stock prices of industry rivals.*

We further explore the potential mechanisms through which ES scandals impact industry competitors. First, if the common risk hypothesis is valid, firms' ES activities could alleviate investor risk apprehensions. Existing literature indicates that firms with high ESG ratings have a lower likelihood of ESG scandals (Sun et al., 2022). Conversely, firms generating more pollution are prone to more regulatory violations (Chatterji et al., 2009). It is, hence, plausible to anticipate that competitors with strong ES performance are likely to exhibit lower ES risk compared to those with weak ES performance. Further, stocks of socially responsible firms, which have lower litigation risk (Koh et al., 2014), default risk (Shih et al., 2021), and stock price crash risk (Kim et al., 2014), are often viewed as insurance by investors who remain more loyal to them, particularly during a negative shock (Albuquerque et al., 2020). Competitors with strong ES performance are, thus, expected to yield higher stock returns than those with weak ES performance during ES scandal announcements. Firms committed to sustainable development also accrue moral capital (Godfrey, 2005; Godfrey et al., 2009) and stakeholder trust (Lins et al., 2017). For instance, ES firms are likely to retain a more loyal customer base and

consequently higher profitability than other firms (Albuquerque et al., 2019), which could also contribute to the resilience of ES firms' stock. Therefore, firms with high ES ratings are anticipated to demonstrate greater resilience during ES scandal announcements. To examine this hypothesis, we propose the following:

H2. *The negative impact of ES scandals is less pronounced for rivals with high ES ratings.*

Moreover, a firm's state ownership could affect investors' apprehensions about ES risk. As a flexible instrument of the government, SOEs are established to address market failures and externalities, such as employment and environmental issues (Hsu et al., 2023). The CFA Institute (2017) asserts that 'SOEs in China are tasked to pursue public interest with regard to environmental engagement'. Additionally, SOEs have superior access to external finance through state-owned banks and credit markets (Cull et al., 2015; Firth et al., 2008), thus having ample cash reserves to undertake ES activities. If investors divest from competitors' stocks due to risk similar to that of the event firms, SOEs are expected to exhibit greater resilience during ES scandal announcements. However, in instances of scandals involving SOEs, the state ownership of competitors might become less advantageous as investors could question or alter their inherent belief that SOEs excel in ES. Hence, we anticipate the resilience of state-owned competitors when the scandal firms are not state-owned. Thus, we hypothesise the following:

H3. *The negative impact of ES scandals on industry rivals is less pronounced when rivals are state-owned and the scandal firms are not state-owned.*

The external governance environment can influence firms' behaviours. With weak external governance, firms face fewer external constraints. They are inclined to imitate each other's detrimental practices and suffer from similar failures in regions (Yu et al., 2015). Consequently, investors may perceive industry rivals as having as high an ES risk as scandal firms when both are in a weak external governance environment. A weak external environment can also reduce firm transparency and increase information asymmetry (Morck et al., 2000). Investors would then rely more on information revealed by event firms to assess the performance of rivals. Therefore, we hypothesise that the contagion effect is stronger when both scandal firms and their industry rivals are in regions with weak external governance. We propose the following hypothesis to test this conjecture:

H4. *The negative impact of ES scandals on industry rivals is more pronounced when both scandal firms and rivals are located in regions with weak external governance.*

Industry competition can also influence the impact of ES scandals on industry rivals. ES scandals might provide beneficial information to industry rivals in low-competition industries, given a demand shift from scandal firms to rivals (Lang & Stulz, 1992), resulting in a competitive effect. Conversely, firms in less competitive industries are more likely to engage in collusive behaviours (Sudharshan, 1995) and share similar operational practices, leading investors to perceive high ES risks among these rivals. Accordingly, ES scandals might convey detrimental news to rivals with a low degree of competition, resulting in a contagion effect. In equilibrium, the influence of industry competition on the rivals depends on which effect dominates the other, which is an empirical research question. However, if the contagion effect is more pronounced in low-competition industries, we can further confirm the common risk argument. We hypothesise the following to examine the impact of industry competition on the industry effect of ES scandals.

H5. *The negative impact of ES scandals on industry rivals is more pronounced in low-competition industries.*

3. Data and methods

3.1. Data

We employ a unique and comprehensive ESG dataset from the IIGF at the Central University of Finance and Economics (CUFE),¹ based on a localised ESG evaluation system² in China. The IIGF uses publicly available information, collected manually and through artificial intelligence. Key sources include firms' ESG, CSR, and annual reports, from which relevant information about firms' ESG processes and performance is extracted. Other significant sources are news releases and penalty information disclosed by governments. Specifically, the IIGF collates negative news and reports on illegal activities of firms from government websites (e.g., the Ministry of Ecology and Environment of China) and Baidu News, cross-verifying the information for accuracy and reliability. Information deemed relevant to ESG is recorded and classified by IIGF analysts based on predetermined criteria (Table A1). For instance, negative ES events encompass various issues such as environmental penalties, illegal emissions, pollution, mass layoffs, salary arrears, strikes, suicides, employee disputes, infringements, plagiarism, contract disputes, customer complaints, production accidents, product quality problems, false advertising, and unlicensed construction.

To quantify ESG performance, the IIGF has developed a comprehensive ESG scoring form comprising three primary indicators, 25 secondary indicators, and over 300 tertiary indicators. It includes both quantitative and qualitative indicators. For qualitative indicators, a set of binary variables is defined, which are either positive (score 1) or neutral (score 0). Under three primary indicators, all qualitative indicators are first aggregated and then normalised based on industry averages. Each quantitative indicator is also normalised to the industry average and can be positive, zero, or negative. The IIGF then constructs a net environmental (social, governance) pillar score by adding the aggregated qualitative and positive quantitative indicators and subtracting negative quantitative indicators. To avoid capturing governance effects, we utilize ES ratings sourced from the IIGF ESG database, which aggregates environmental and social metrics. Additional information, including stock returns, firm size, address, financial leverage, state ownership, largest shareholder's percentage ownership, Herfindahl Index (HHI), and regional GDP per capita, is sourced from the China Stock Market & Accounting Research Database (CSMAR).³

3.2. Sample selection

This study examines corporate ES scandals involving A-share listed

¹ The IIGF is China's premier international green finance research institute. It collaborates with various national, regional, and local government institutions, financial institutions, and research organisations, including the People's Bank of China, the Chinese Ministry of Finance, the National Development and Reform Commission, and the Chinese Ministry of Environmental Protection, to advance green finance in China. On an international scale, IIGF undertakes numerous collaborative green finance research projects with organisations such as UNEP, UN PRI, the European Investment Bank, Cambridge University, and the International Institute for Sustainable Development. Further information about IIGF can be found at <http://iigf.cufe.edu.cn/>.

² Utilising the IIGF ESG evaluation system and database, several indices have been compiled, including the CSI CUFE SH-SZ 100 ESG Leading Index, CSI300 Green Leading Stock Index, CUFE-CNI SZ-HK Connect Green Selection Index, etc. In the international capital markets, the IIGF ESG database is available on Deutsche Borse, marking it as the first Chinese ESG database to be included. Furthermore, IIGF and Qontigo Indices have jointly developed the STOXX-IIGF China A-share ESG Index, the first international ESG index based on a Chinese ESG database.

³ CSMAR is a leading financial data provider in China, widely used in literature to extract corporate fundamental information. Examples of such usage include Yu et al. (2015), Cui et al. (2018), and Shan and Tang (2023).

companies in China from 2014 to 2021. Data, including company ticker, name, event date (i.e., the date a scandal occurred or became public), event description, and issue area (environment or social), are sourced from the IIGF ESG database. We exclude corporate governance scandals, following Kruger (2015), for two reasons. Firstly, enhancing corporate governance does not necessitate financial expenditure, unlike the other two areas. Secondly, the inclusion of shareholders as stakeholders in ESG is debatable. By excluding corporate governance scandals, the focus shifts to firms' non-shareholder stakeholders.

To mitigate the impact of confounding events, firms experiencing major events (e.g., mergers and acquisitions, delisting risk warnings, dividend changes, CEO turnover, IPOs, etc.) within 10 trading days before or after ES scandals are excluded. If a firm has multiple events within a year, only the first event is included in our sample, as subsequent disclosures are less informative. Following Yu et al. (2015), events with positive abnormal returns are also excluded to concentrate on news conveying negative information to event firms. Two potential reasons for negative news not resulting in a drop in stock returns are either investor inattention or failure to perceive them as scandals. To analyse the potential industry-wide effect more effectively, the study focuses solely on ES scandals that elicit negative market reactions.⁴

The final sample encompasses 615 ES scandals, comprising 248 environmental and 367 social events. Industries are identified using a historical, three-digit classification code as per the *Industry Classification Guidance of Listed Companies* issued by the China Securities Regulatory Commission in 2012.⁵ This process yields a rival sample consisting of 46,831 firm-event observations with adequate stock return and financial data. As depicted in Panel A of Table 1, ES scandals have seen a marked increase since 2018. We attribute this not to a sudden surge in ES scandals, but to heightened disclosure driven by growing public interest in corporate sustainability. Panel B reveals that our sample spans 71 industries, with negative environmental incidents reported in 47 industries and social incidents in 62. The chemical and pharmaceutical industries appear most vulnerable, accounting for approximately a third of total scandals.

3.3. Methodology

We employ the event study methodology⁶ to gauge the stock market's reaction to 615 ES scandals, considering the implicated firms and their industry competitors. We designate the day the market first learns of an ES scandal as the event day (day 0). If this is not a trading day, the first trading day following the announcement becomes the event day. As markets may receive information in advance given leaks, we set the event window prior to the event day (e.g., 2 days). Daily abnormal returns (AR) are computed as residuals from the market model, using parameters estimated from the 220 to 11 days preceding the scandals.⁷

⁴ To ensure robustness, we also investigate the industry effect for all events, including those with positive market reactions. The findings, although not reported here, align with our main results.

⁵ The *Industry Classification Guidance of Listed Companies*, an official and mainstream industry classification criterion used by institutional investors in China, comprises 90 industries at its most detailed level. For instance, B07 denotes the oil and natural gas exploitation industry, while C28 signifies the chemical fibre manufacturing industry.

⁶ The event study methodology, initially introduced by Dolley (1933) and later developed by Ball and Brown (1968) and Fama et al. (1969), is commonly used to determine stock market reactions. This method has been adopted in numerous studies, such as Endrikat (2016), Hertzel et al. (2008), Lang and Stulz (1992), Slovin et al. (1991), and Xu et al. (2012).

⁷ AR is the difference between the realised return and the expected return. The formula is as follows: $AR_{it} = R_{it} - \hat{R}_{it}$, where R_{it} is the realised return and $\hat{R}_{it}$ is the expected return. The expected return is estimated using a market model. The formula is as follows: $\hat{R}_{it} = \hat{\alpha} + \hat{\beta}R_{mt} + \varepsilon_{it}$, where $\hat{\alpha}$ and $\hat{\beta}$ are estimated from ordinary least squares regression between 220 and 11 days prior to the scandals. R_{mt} represents the market return on day t .

Table 1

Sample distribution of environmental and social (ES) scandals.

Panel A: Sample distribution by year			
Year	All scandals	Environmental scandals	Social scandals
2014	42	16	26
2015	34	20	14
2016	44	17	27
2017	59	19	40
2018	107	41	66
2019	114	28	86
2020	127	71	56
2021	88	36	52
Total	615	248	367

Panel B: Sample distribution by industry			
Industry	All scandals	Environmental scandals	Social scandals
Agriculture	2	1	1
Forestry	1	0	1
Animal husbandry	2	2	0
Coal mining and dressing	11	1	10
Oil and natural gas exploitation	4	1	3
Ferrous metal ore mining and dressing	1	0	1
Non-ferrous metal ore mining and dressing	11	1	10
Non-metallic ore mining and dressing	2	0	2
Exploitation auxiliary activities	1	0	1
Agricultural and sideline food processing	8	6	2
Food manufacturing	12	3	9
Alcohol, beverage and refined tea manufacturing	13	4	9
Textile industry	4	0	4
Textile garment and apparel	8	0	8
Leathers, furs, feather products and footwear	3	1	2
Wood processing and wood, bamboo, rattan, palm fibre, and straw products	1	0	1
Furniture manufacturing	5	2	3
Papermaking and paper products	6	6	0
Printing and recording media reproduction	1	1	0
Stationery, industrial arts, sports and entertainment supplies manufacturing	2	0	2
Petroleum, nuclear fuel processing and coking	7	4	3
Chemical raw materials and products manufacturing	84	50	34
Chemical fibre manufacturing	5	1	4
Pharmaceutical industry	62	31	31
Rubber and plastic products	8	5	3
Non-metallic mineral products	15	9	6
Ferrous metal smelting and rolling processing	17	8	9
Non-ferrous metal smelting and rolling pressing	23	14	9
Metal products	8	3	5
General equipment manufacturing	11	5	6
Special-purpose equipment manufacturing	27	7	20
Automobile manufacturing	21	6	15
Railways, ships, aircraft, spacecraft manufacturing	6	4	2
Electric machinery and equipment manufacturing	15	4	11
Computer and communications manufacturing	27	12	15
Instrument and metre manufacturing	2	1	1
Other manufacturing industries	3	3	0
Comprehensive utilisation of waste resources	2	1	1

(continued on next page)

Table 1 (continued)

Panel B: Sample distribution by industry			
Industry	All scandals	Environmental scandals	Social scandals
Electric power and heat production and supply	9	7	2
Water production and supply	4	4	0
Building construction	1	1	0
Civil engineering construction	18	8	10
Architectural decoration and other construction	4	0	4
Wholesale	7	2	5
Retail industry	15	1	14
Railway transportation	2	1	1
Road transport industry	4	2	2
Waterway transport	8	6	2
Loading/unloading handling and transport agency	1	1	0
Storage industry	1	0	1
Postal service	4	0	4
Accommodation industry	1	1	0
Catering industry	2	1	1
Telecommunications, radio and television, and satellite transmission services	1	0	1
Internet and related services	8	0	8
Software and information technology services	9	0	9
Monetary and financial services	8	1	7
Capital market services	4	0	4
Insurance industry	1	0	1
Other financial industries	1	0	1
Real estate	30	6	24
Leasing industry	1	0	1
Commercial service	2	0	2
Research and experimental development	1	0	1
Professional technical service	2	0	2
Ecological protection and environmental governance	7	7	0
Health	5	0	5
Press and publishing industry	1	0	1
Radio, television, film, and film and television sound recording production	6	1	5
Culture and arts	3	0	3
Diversified industries	3	1	2
Total	615	248	367

Note: This table provides a summary of the distribution of ES scandals among firms listed on the A-share markets of the Shanghai Stock Exchange and Shenzhen Stock Exchange in China from 2014 to 2021. Industries are classified based on a historical three-digit code, as per the *Industry Classification Guidance for Listed Companies* issued by the China Securities Regulatory Commission in 2012.

We then assess cumulative abnormal return (CAR) by summing up AR over multiple days, constituting the event window.⁸ To examine the overall market reaction to multiple events, we calculate cumulative average abnormal return (CAAR) as the average CARs.⁹ Additionally, we employ the three-factor model (Fama & French, 1993) and four-factor model (Carhart, 1997) to verify the robustness of our findings.

To probe potential determinants of information transfer, we proceed with conventional regression analyses. Since ES scandals are external to industry rivals, we can feasibly alleviate concerns about reverse causality and scrutinise the causal relationship from ES to stock performance. We also correlate rivals' stock returns with their ES risk in the preceding year to protect against the chance that rivals alter their ES

policies in reaction to unexpected industry scandals. We investigate the ensuing cross-sectional variation using ordinary least squares regression, a method proven suitable for analysing 'event clustering' cases (Karafiath, 2009). We incorporate year and industry-fixed effects to control for certain unobservable effects and cluster standard errors by firm. The formula is as follows:

$$CAR\ rival_i = \alpha + \beta ES\ risk_i + \delta X_i + Year\ dummies + Industry\ dummies + \varepsilon_i \quad (1)$$

where $CAR\ rival_i$ denotes rival firms' CARs in the event window $(-2, 2)$. To investigate the influence of rivals' ES risk on the contagion stock returns, we incorporate a vector of variables, symbolised by $ES\ risk_i$. This includes $ES\ rival_i$, derived from ES ratings in the IIGF ESG database, signifying the ES performance of rival firms. $SOE\ rival_i$ is an indicator that equals 1 for state-owned rivals and 0 for non-state-owned rivals. A firm is state-owned if the government is its ultimate controlling shareholder. $Spillovers\ (nsoe\ to\ soe)_i$ is an indicator that equals 1 if an event firm is a non-SOE while its rival firm is an SOE, and 0 otherwise. $Spillovers\ (soe\ to\ soe)_i$ is an indicator that equals 1 if an event firm and its industry rival are SOEs, and 0 otherwise. $Legal_i$ and GDP_i capture a firm's external environment. Specifically, $Legal_i$ equals 1 if an event firm and its industry rival are in regions with weak legal protection, and 0 otherwise. The regional legal protection is gauged with the Legal Environment Index in the National Economic Research Institution Index of Marketization (IM).¹⁰ The higher the value, the stronger the legal protection of a region. We categorise provinces into regions with weak and strong legal protection based on the median Legal Environment Index value. GDP_i equals 1 if an event firm and its industry rival are located in less developed regions, and 0 otherwise. We divide the provinces into less and more developed regions based on the median of GDP per capita. HHI_i denotes the degree of industry competition. It equals 1 if a rival firm is in a low-competition industry and 0 if it is in a high-competition industry. We divide the industries into high- and low-competition industries based on the median of the Herfindahl index.

We select some variables that may influence rivals' ARs as our control variables, denoted by X_i . Specifically, similar to prior studies (Xu et al., 2006; Yu et al., 2015), control variables include the following: $ES\ event_i$, which is event firms' ES ratings in the IIGF ESG database in the previous year; $CAR\ event_i$, which is the five-day AR of event firms, capturing the magnitude of information conveyed by event firms; $Size\ event_i$, which is the logarithm of event firms' total assets at the end of the previous year; $Size\ rival_i$, which is the logarithm of rival firms' total assets at the end of the previous year; $Leverage_i$, rival firms' financial leverage ratio, which is the ratio of total liabilities to total assets at the end of the previous year; Top_i , which is rival firms' largest shareholder's ownership at the end of the previous year; and $Similar_i$, which is the correlation of stock returns between event firms and rivals from 220 to 11 trading days before the event day. Table A2 in the Appendix presents more detailed variable definitions.

4. Industry effect of environmental and social scandals

4.1. Stock price responses of industry rivals

To examine the industry impact of ESG scandals, we analyse the stock returns of rival firms during these events. Using a historical three-digit industry classification code for matching, we compile a rival sample comprising 46,831 firm-event observations. Fig. 1 illustrates the market's response to individual rivals over a 51-day period surrounding the scandal announcements. Despite an overall negative and declining trend in the CARs of the rivals, the decline is not specifically

⁸ The formula is as follows: $CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{it}$, where (t_1, t_2) represents the event window.

⁹ The formula is as follows: $CAAR(t_1, t_2) = \frac{1}{N} \sum_{i=1}^N CAR_i(t_1, t_2)$, where N represents the number of firms.

¹⁰ This index is frequently employed in China-based studies, such as those by Fan et al. (2013), Wang et al. (2008), and Yu et al. (2015).

concentrated on event days. This could be attributed to the industry effect being dispersed due to the sufficient diversification of rival firms (Xu et al., 2006; Zou et al., 2015). Alternatively, the results could be skewed by certain industries with a higher number of rivals. For instance, the chemical, pharmaceutical, and computer and communications industries each have approximately 200 industry rivals, accounting for over half of the total sample. Conversely, industries (e.g., postal, catering, and accommodation services) have no more than three rivals.

To mitigate this dispersion, we further estimate the industry effect of ES scandals by capping the maximum number of rival firms for each scandal, following the approach used by Xu et al. (2006) and Zou et al. (2015). We examine 50 rivals sharing the same three-digit industry classification code as the event firms and whose cash flow characteristics most closely resemble those of the event firms. If an event firm has fewer than 50 rivals, all available rivals are included in our sample. In line with Lang and Stulz (1992) and Xu et al. (2006), we use the pre-announcement stock return correlation between event firms and industry rivals to represent cash flow similarity.¹¹ This results in a rival sample of 23,051 firm-event observations.

Fig. 2 depicts the trend of cumulative average ARs for the revised rival sample. On average, rival firms did not exhibit persistent negative ARs until four days prior to the scandal announcements. This aligns with our expectations, as ES scandals typically remain undisclosed or receive minimal investor attention until they are made public. For instance, it is improbable that markets would receive pertinent information before events such as explosions or staff suicides occur. From day -4 to day 17, rivals experience a significant decline in stock returns, with an average CAR of 1.1% over the 22 trading days. Rivals are adversely affected by ES scandals for approximately four weeks, likely due to sustained media coverage. The rivals then cease to exhibit persistent negative ARs. The empirical results indicate that the contagion effect supersedes the competitive effect upon the announcement of ES scandals, suggesting that investors tend to attribute ES scandals to industry risks rather than idiosyncratic (firm-specific) risks.

The analysis continues with detailed estimates of market reactions to rival firms. As depicted in Panel A of Table 2, the average shareholder wealth loss for rivals is 0.43% within the $(-2, 2)$ window. This loss appears significantly smaller than that of the event firms (4.99%). However, considering the market value, rivals' total losses amount to 1.43 billion yuan, exceeding the loss of event firms (0.89 billion yuan) by more than 1.5 times. Consequently, ES scandals exert a substantial impact on the entire industry. To explore the medium- and long-term effects of ES scandals, we establish two post-announcement windows. The findings reveal that scandals persistently exert a significantly negative impact on industry rivals, but not on event firms, from 3 to 20 days post-announcement. In the long run, ES scandals do not significantly influence either event firms or their industry rivals.

4.2. Robustness tests

This section validates the robustness of our primary findings using various alternative outcome measures and rival samples. We first group each event firm's rivals into a portfolio, treating the portfolio as a distinct observation, to ascertain whether our findings are influenced by outcome measurement errors. Using the market model, we compute the stock returns of both value- and equal-weighted portfolios.¹² As depicted in Panel A of Table 3, the stock market exhibits a significant negative

response to the rival portfolios. The average percentage wealth loss in the $(-2, 2)$ window varies from 0.63% to 0.69%, a magnitude even larger than the loss associated with individual rivals. We further validate our results using alternative measures of stock market response. Specifically, we recompute the ARs based on the three-factor model (Fama & French, 1993) and the four-factor model (Carhart, 1997). The results, presented in Panel B of Table 3, affirm our primary conclusion that ES scandals impart negative information to industry rivals on average.

To address potential bias in our results due to the selection of rival samples, we estimate with an alternative set of rival firms. We rematch the industry rivals using the CITIC Industry Classification Standard, another prevalent industry classification criterion among institutional investors in China. Table 4 shows significant and negative ARs for rivals around the scandal announcements, reinforcing the contagion effect of ES scandals.

To dispel concerns that our findings may be influenced by the number of rivals we designate, we reassess the industry effect of ES scandals with several predetermined numbers of rivals. We examine similar rivals by capping the maximum number of rivals to 5, 25, 75, and 100 for each event firm. As shown in Table 5, the results align with our main findings in terms of statistical and economic significance. The contagion effect appears to be driven by the top 5 similar rivals. We then recompute the ARs excluding these top 5 rivals for the subsequent 20, 70, and 95 similar rivals. Our unreported results suggest that the negative stock returns are not driven by the top 5 rivals.

Cao et al. (2019) suggest that other confounding factors, such as market sentiments, might influence the results. To address this, we conduct a placebo test by simulating 'incorrect' industry rivals. We replace each rival with a pseudo-rival, ensuring the two firms belong to different industries and share the same decile group in terms of size, leverage, return on assets, and book-to-market ratio. If more than one pseudo-rival is found, we retain the one with the firm leverage closest to that of the rival firm. Since the pseudo-rivals are not the industry rivals of event firms, we anticipate no significant reactions by the pseudo-rivals to the announcement of ES scandals. As in Table 6, the average AR of the pseudo-rivals is positive but insignificant on the event day. Furthermore, the ARs in the two days following the scandals and the five-day CARs are also insignificant. Thus, ES scandals do not exert a contagion effect on the pseudo-rivals and our results are robust.

4.3. Cross-sectional analyses

We first investigate the industry effects for environmental- and social-related scandals separately. As indicated in Panel A of Table 7, rival firms experience a significant and negative average AR over the period $(-2, 2)$ in both types of scandals. The contagion effect is more pronounced in environmental scandals. Specifically, the five-day CAR for rivals implicated in environmental scandals is -0.72% , considerably lower than the -0.22% AR for rivals implicated in social scandals. Given the negligible difference in ARs between the two types of event firms, we dismiss the possibility that the difference in industry effects is attributable to differences in scandal impacts. Our findings suggest that investors perceive environmental scandals as a greater industry risk than social scandals. Indeed, approximately 50 A-share listed firms, predominantly in the chemical and pharmaceutical industries, disclosed penalties for water or air pollution in the 2019–2020 period. In contrast, according to the IIGF ESG ratings, employee relations vary significantly among firms within the same industry.

In addition, the nature of the business is a determinant of ES performance (Shih et al., 2021). For instance, industries such as leasing, insurance, and agriculture typically exhibit a smaller negative environmental impact compared to "two-high-and-one-overcapacity" (THOO) industries characterized by high pollution, energy consumption, and excessive productivity. Equity investors may have already factored ES risks into the equity prices of THOO firms. Therefore, ES scandals in these industries may not surprise investors, resulting in a

¹¹ We calculate the cash flow similarity using the correlation of stock returns between event firms and industry rivals, measured from 220 to 11 trading days before the event day.

¹² For value-weighted portfolios, daily portfolio returns are calculated using value weights derived from firms' market values. In contrast, for equal-weighted portfolios, portfolio returns are estimated using equal weights.

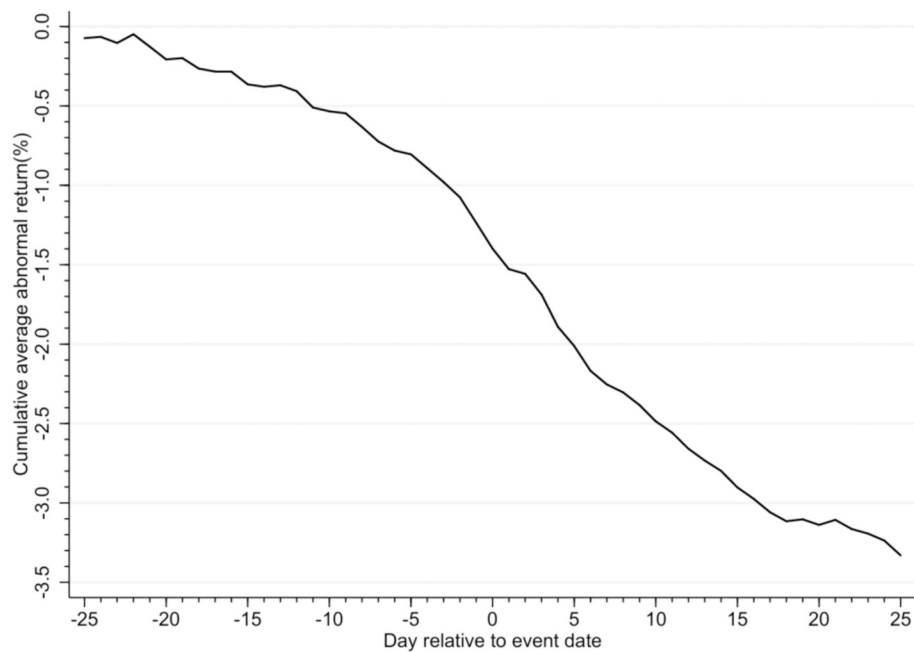


Fig. 1. Cumulative average abnormal returns of industry rivals.

Note: This figure plots the cumulative average abnormal returns of all industry rivals over a 51-day period surrounding ES scandals. Industry rivals are matched using a historical, three-digit industry classification code as per the Industry Classification Guidance of Listed Companies, issued by the China Securities Regulatory Commission in 2012.

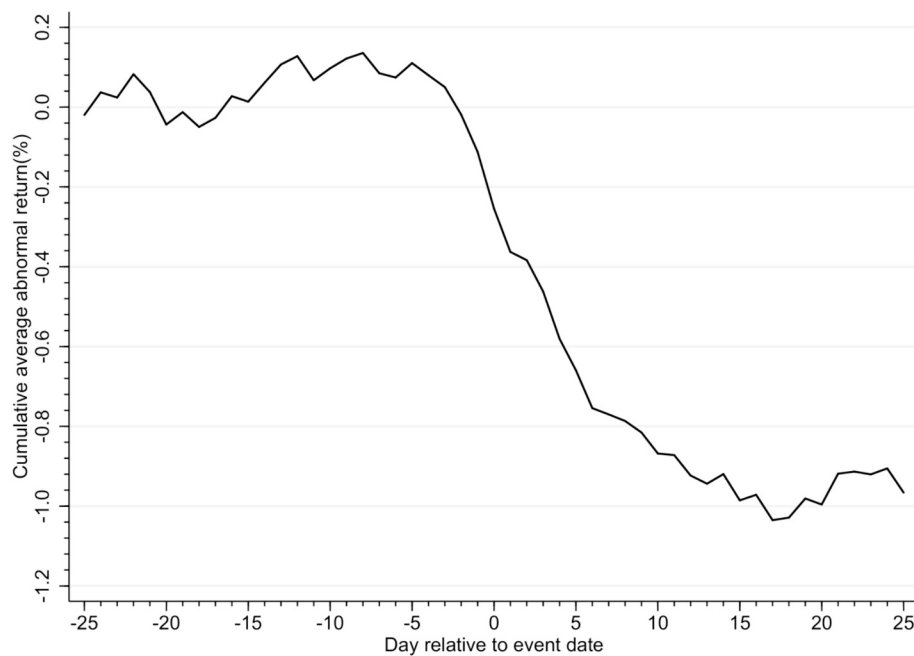


Fig. 2. Cumulative average abnormal returns of similar industry rivals.

Note: This figure plots the cumulative average abnormal returns for similar industry rivals over a 51-day period surrounding ES scandals. Each scandal includes up to 50 rivals that share the same three-digit industry classification code as the event firms and exhibit cash flow characteristics most similar to those of the event firms. Cash flow similarity is determined by the correlation of stock returns between event firms and industry rivals from 220 to 11 trading days prior to the event day.

limited contagion effect. However, such scandals could prompt investors to divest from both the affected firms and their competitors in *THOO* industries to manage risk, potentially amplifying contagion effects. To examine the heterogeneity of industry effects, we compare CARs between *THOO* industries and others. As presented in Panel B of Table 7, the contagion effect is less pronounced in *THOO* industries compared to other sectors, supporting the former argument.

Further, in 2016, the People's Bank of China, in conjunction with six other governmental agencies, issued the *Guidelines for Establishing the Green Financial System*. This initiative has heightened societal awareness of sustainable practices, prompting more firms to disclose ESG reports. As previously discussed, information asymmetry can exacerbate industry effects, particularly contagion. Increased disclosure of ESG information helps alleviate information asymmetry, thereby potentially

Table 2

Market reactions of events firms and industry rivals.

Windows	Event firms			Industry rivals		
	N	AR	t-stat.	N	AR	t-stat.
−2	615	−0.67***	−7.34	23,051	−0.07***	−4.51
−1	615	−0.96***	−9.49	23,051	−0.09***	−6.04
0	615	−1.48***	−12.81	23,051	−0.14***	−9.43
1	615	−1.14***	−12.01	23,051	−0.11***	−6.95
2	615	−0.74***	−7.96	23,051	−0.02	−1.31
(−2, +2)	615	−4.99***	−10.79	23,051	−0.43***	−5.47
(−20, −3)	615	−0.76	−0.41	23,051	0.01	0.05
(+3, +20)	615	−1.38	−0.75	23,051	−0.61**	−2.04
(+21, +30)	615	−0.43	−0.39	23,050	0.02	0.12

Note: The table presents the market reactions of event firms and their industry rivals in response to ES scandals. The abnormal return is estimated using the standard market model, with parameters derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed *t*-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean abnormal return (cumulative abnormal return) differs from zero.

Table 3

Alternative measures of market reactions of industry rivals.

Panel A: Rivals are formed with portfolios						
Windows	Value-weighted portfolios			Equal-weighted portfolios		
	N	AR	t-stat.	N	AR	t-stat.
−2	612	−0.11**	−2.23	612	−0.12**	−2.55
−1	612	−0.17***	−3.73	612	−0.14***	−3.20
0	612	−0.18***	−3.90	612	−0.20***	−4.75
1	612	−0.08*	−1.74	612	−0.14***	−3.34
2	612	−0.09*	−1.75	612	−0.09*	−1.94
(−2, +2)	612	−0.63**	−2.53	612	−0.69***	−2.97
(−20, −3)	612	0.27	0.31	612	−0.15	−0.18
(+3, +20)	612	0.12	0.12	612	−0.66	−0.71
(+21, +30)	612	0.34	0.62	612	0.09	0.16

Panel B: ARs (CARs) are estimated with the three- and four-factor model						
Windows	The three-factor model			The four-factor model		
	N	AR	t-stat.	N	AR	t-stat.
−2	23,051	−0.03**	−2.12	23,051	−0.02	−1.08
−1	23,051	−0.07***	−5.00	23,051	−0.08***	−5.02
0	23,051	−0.12***	−8.28	23,051	−0.13***	−8.63
1	23,051	−0.09***	−5.99	23,051	−0.09***	−5.53
2	23,051	0.02	1.24	23,051	0.01	0.10
(−2, +2)	23,051	−0.30***	−3.81	23,051	−0.30***	−3.82
(−20, −3)	23,051	−0.20	−0.75	23,051	−0.14	−0.54
(+3, +20)	23,051	−0.07	−0.24	23,051	−0.05	−0.17
(+21, +30)	22,799	−0.07	−0.40	22,799	−0.04	−0.23

Note: The table outlines alternative measures of market reactions to industry rivals. In Panel A, all rivals are treated as a portfolio for each event firm, and the abnormal return is estimated using the standard market model. In Panel B, the abnormal return is estimated using the Fama-French three-factor model (Fama & French, 1993) and the Carhart four-factor model (Carhart, 1997). For all models, parameters are derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed *t*-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean abnormal return (cumulative abnormal return) differs from zero.

mitigating the contagion impact of ES scandals. Conversely, heightened public attention on ES issues may attract media scrutiny, facilitating the exposure of ES misconduct among rivals and intensifying contagion effects. Panel C in Table 7 demonstrates a diminished contagion effect of

Table 4

Alternative industry classification.

Windows	Abnormal returns for rivals			
	N	AR	t-stat.	
−2	10,271	−0.14***	−6.15	
−1	10,271	−0.13***	−5.93	
0	10,271	−0.19***	−8.53	
1	10,271	−0.15***	−6.66	
2	10,271	−0.11***	−4.72	
(−2, +2)	10,271	−0.71***	−6.31	
(−20, −3)	10,271	−0.48	−1.18	
(+3, +20)	10,271	−1.06**	−2.46	
(+21, +30)	10,271	−0.45*	−1.81	

Note: In this table, industry rivals are redefined according to the CITIC Industry Classification Standard. The abnormal return is estimated using the standard market model, with parameters derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed *t*-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean abnormal return (cumulative abnormal return) differs from zero.

Table 5

Arbitrarily assigned numbers of rivals.

Panel A: Abnormal returns for 5 and 25 rivals						
Windows	Abnormal returns for 5 rivals			Abnormal returns for 25 rivals		
	N	AR	t-stat.	N	AR	t-stat.
−2	3008	−0.07*	−1.70	13,484	−0.07***	−3.40
−1	3008	−0.14***	−3.50	13,484	−0.09***	−4.64
0	3008	−0.19***	−4.80	13,484	−0.14***	−7.33
1	3008	−0.08*	−1.88	13,484	−0.11***	−5.48
2	3008	−0.06	−1.54	13,484	−0.05**	−2.57
(−2, +2)	3008	−0.53***	−2.60	13,484	−0.45***	−4.59
(−20, −3)	3008	0.54	0.76	13,484	0.24	0.68
(+3, +20)	3008	−0.16	−0.21	13,484	−0.30	−0.79
(+21, +30)	3008	0.31	0.72	13,484	0.17	0.77

Panel B: Abnormal returns for 75 and 100 rivals						
Windows	Abnormal returns for 75 rivals			Abnormal returns for 100 rivals		
	N	AR	t-stat.	N	AR	t-stat.
−2	30,245	−0.07***	−5.01	36,049	−0.07***	−5.69
−1	30,245	−0.09***	−6.97	36,049	−0.11***	−8.64
0	30,245	−0.13***	−9.79	36,049	−0.14***	−11.16
1	30,245	−0.11***	−7.70	36,049	−0.11***	−8.86
2	30,245	−0.02	−1.37	36,049	−0.02	−1.21
(−2, +2)	30,245	−0.42***	−6.04	36,049	−0.45***	−6.97
(−20, −3)	30,245	−0.12	−0.47	36,049	−0.26	−1.14
(+3, +20)	30,245	−0.78***	−2.95	36,049	−0.96***	−3.91
(+21, +30)	30,245	−0.04	−0.27	36,049	−0.12	−0.89

Note: This table presents a scenario where 5, 25, 75, and 100 rivals are arbitrarily assigned to each event, respectively. Samples are selected based on their cash flow characteristics, which are most similar to those of the event firms. The abnormal return is estimated using the standard market model, with parameters derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed *t*-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean abnormal return (cumulative abnormal return) differs from zero.

ES scandals following the issuance of the guidelines, affirming the notion that reduced information asymmetry can attenuate the adverse influence of ES scandals on industry peers.

5. Mechanism analyses

Prior research establishes that ES scandals exert a significant,

Table 6
Placebo test.

Windows	Abnormal returns for rivals		
	N	AR	t-stat.
-2	6343	-0.13***	-4.66
-1	6343	-0.08**	-2.52
0	6343	0.03	0.89
1	6343	-0.05	-1.65
2	6343	-0.01	-0.39
(-2, +2)	6343	-0.24	-1.59

Note: This table presents a placebo test with matched pseudo-rivals. For each industry rival, a pseudo-rival is identified based on firm size, leverage, return on assets, and book-to-market ratio. The abnormal return is estimated using the standard market model, with parameters derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed t-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean abnormal return (cumulative abnormal return) differs from zero.

negative impact on industry competitors. This suggests that investors may attribute ES scandals to risks common across the industry, rather than risks specific to individual firms. In this section, we further explore the mechanism underlying this contagion effect.

5.1. Impact of industry rivals' environmental and social performance on contagion stock returns

5.1.1. Environmental and social performance and contagion stock returns

We proceed with conventional regression analyses to examine the impacts of industry rivals' ES performance on contagion stock returns. Table 8 provides the summary statistics of the variables used in the regression tests. In the five days surrounding ES scandal announcements, rivals' stock prices decline by a mean (median) AR of -0.43% (-0.62%). The mean level of rivals' ES rating is 0.45, with a standard deviation of 0.31, indicating significant variation in ES performance across rival firms. Further, we also conduct a univariate analysis to compare the ES performance of firms with and without an ES scandal. As depicted in Fig. 3, firms with ES scandals have significantly lower ES ratings than those without ES scandals between 2014 and 2021. This evidence suggests that firms lagging in ES are more likely to have ES scandals, as in Sun et al. (2022).

To isolate the effect of industry rivals' ES performance on contagion stock returns, we apply the regression model (1). As demonstrated in column (1) of Table 9, the coefficient of $ES\ rival_i$ is positive and statistically significant at the 1% level, using unrestricted regression models.¹³ The magnitude of the coefficient estimate implies that a one standard deviation increase in rivals' ES ratings correlates with an average AR increase of 0.15% (0.474×0.311). This increase equates to approximately 35% ($0.15/0.43$) of the industry rivals' contagious stock returns in the five days surrounding the scandals, a non-trivial amount. This empirical evidence aligns with our expectation that good ES performance from rivals can mitigate the negative impact of industry ES scandals, suggesting, as in Hoepner et al. (2022), a risk reduction effect of ES activities. Regarding the control variable coefficient estimates, the significant and positive coefficient of $CAR\ event_i$ confirms the information spillover effect of scandals to their industry rivals. The significant and negative coefficient of $Similar_i$ shows that the event firm and

industry rivals' cash flow similarity will amplify ES scandals' contagion effect, as per Lang and Stulz (1992). In addition, we use an alternative ES performance measure¹⁴ and an alternative event window¹⁵ to check the robustness of our findings and find consistent results.

5.1.2. Environmental and social performance, information disclosure, and contagion stock returns

ES information disclosure enables investors to reduce information asymmetry (Siew et al., 2016) and accurately evaluate firms' performance. Based on this, we predict that if rivals with high ES ratings disclose ES information through ESG reports, they are more likely to be acknowledged by investors for their ES ratings and consequently have higher returns during scandals. However, for rivals with low ES ratings, disclosure is unlikely to be beneficial, as investors may suspect them of greenwashing due to a discrepancy between their claims and actions.

To examine the effect of ES information disclosure on the relationship between ES performance and returns during scandals, we replicate the previous analysis, allowing the effect of ES on the five-day ARs to differ based on whether the industry rivals disclose ESG reports. As shown in column (2) of Table 9, the coefficient of the interaction term $ES\ rival_i \times Disclosure\ rival_i$ is positive and statistically significant at the conventional level, indicating that the positive correlation between ES and stock returns is more pronounced for rivals who disclose ESG reports compared to those who do not. Moreover, the coefficient of $Disclosure\ rival_i$ is significantly negative, indicating that the disclosure of ESG reports exacerbates the negative impact of ES scandals for rivals with low ES ratings. The empirical evidence validates our hypothesis that the positive relationship between ES and stock returns during scandals operates through the risk channel.

5.2. Further analyses

5.2.1. Impact of state ownership on contagion stock returns

In China, firms have substantial state ownership,¹⁶ providing an opportunity to examine the impact of state ownership on the stock returns of rival firms during industry ES scandals. As in column (1) of Table 10, the coefficient of $SOE\ rival_i$ is positive and statistically significant at the 1% level, suggesting that state ownership among rivals can alleviate the contagion effect of ES scandals. In columns (2) and (3) of Table 10, we consider the event firms. The empirical evidence shows that the coefficient of $Spillovers\ (nsoe\ to\ soe)_i$ is significantly positive, indicating a weaker contagion effect when the event firms are non-SOEs and the rivals are SOEs. Conversely, the coefficient of $Spillovers\ (soe\ to\ soe)_i$ is insignificant, suggesting that state ownership among rivals does not mitigate the negative impact of ES scandals when the event firms are SOEs. These findings support Hypothesis 3's argument.

5.2.2. Impact of the external governance environment on contagion stock returns

To deepen our understanding of the external environment's impact on industry rivals' contagious stock returns, we use two proxies for the external governance environment: $Legal_i$ and GDP_i . $Legal_i$ is an indicator

¹³ To ensure comparability of firm size, we also calculate rivals' size as the ratio of the logarithm of rivals' total assets to the logarithm of event firms' total assets at the end of the preceding year. Similar results are obtained when we use relative firm size as a control variable, details of which are available upon request.

¹⁴ For robustness, we repeat the analysis with Sino-Securities ESG ratings, which covers all A-share listed firms since 2009 and is among the mainstream ESG ratings in China. This rating system employs a nine-step scale rather than providing detailed ESG scores. Firms are assigned scores based on their rating levels, with C-rated firms scoring 1 point, CC-rated firms scoring 2 points, and so on, up to AAA-rated firms scoring 9 points. Consistent results are found, details of which are available upon request.

¹⁵ We use a longer event window of the nine-day (-4, 4) CARs for robustness analysis. The findings are consistent, details of which are available upon request.

¹⁶ In China, SOEs play an undeniable role in the economy, accounting for approximately one-third of the total number of listed firms and two-thirds of stock market capitalisation in recent years.

Table 7
Cross-sectional analysis.

Panel A: Environmental vs. social scandals						
Windows	Event firms			Industry rivals		
	E scandals	S scandals	Diff	E scandals	S scandals	Diff
	CAR	CAR	CAR	CAR	CAR	CAR
(−2, +2)	−5.18*** (−6.68)	−4.86*** (−8.53)	−0.32 (−0.71)	−0.72*** (−5.59)	−0.22** (−2.24)	−0.50*** (−6.75)
Panel B: THOO vs. N-THOO industries						
Windows	Event firms			Industry rivals		
	THOO	N-THOO	Diff	THOO	N-THOO	Diff
	CAR	CAR	CAR	CAR	CAR	CAR
(−2, +2)	−4.82*** (−5.85)	−5.06*** (−9.08)	0.24 (0.49)	−0.31** (−2.06)	−0.49*** (−5.28)	0.18** (−2.19)
Panel C: Pre- vs. post-policy						
Windows	Event firms			Industry rivals		
	Pre-policy	Post-policy	Diff	Pre-policy	Post-policy	Diff
	CAR	CAR	CAR	CAR	CAR	CAR
(−2, +2)	−4.53*** (−3.67)	−5.06*** (−10.14)	0.53 (0.81)	−0.65** (−2.42)	−0.40*** (−4.92)	−0.25** (−2.24)

Note: The table presents the results of subsample analyses. In Panel A, we divide the event firms and industry rivals into two groups according to the types of scandals, respectively. In Panel B, we divide the event firms and industry rivals into two groups according to the industry classification, respectively. The two-high-and-one-overcapacity industries are denoted by *THOO* industries, whereas the rest of the industries are denoted by *N-THOO* industries. In Panel C, we divide the event firms and industry rivals into two groups according to the 2016 *Guidelines for Establishing the Green Financial System*, respectively. The abnormal return is estimated using the standard market model, with parameters derived from 220 days to 11 days prior to the event day. The event day (day 0) is defined as the day when the market first receives information about the ES scandals. If this is not a trading day, the first trading day following the announcement is marked as the event day. The symbols *, **, and *** indicate a significant two-tailed t-test at the 10%, 5%, and 1% levels, respectively, with the null hypothesis that the mean cumulative abnormal return differs from zero.

Table 8
Descriptive statistics.

Variable	Obs.	Mean	Std. Dev	25%	Median	75%
<i>CAR rival_i</i>	23,051	−0.434	5.589	−2.969	−0.617	1.828
<i>CAR event_i</i>	23,051	−5.016	5.264	−6.883	−3.236	−1.578
<i>ES rival_i</i>	23,051	0.451	0.311	0.143	0.448	0.732
<i>ES event_i</i>	23,051	0.461	0.326	0.116	0.487	0.772
<i>Disclosure rival_i</i>	23,051	0.273	0.445	0	0	1
<i>SOE rival_i</i>	23,051	0.374	0.484	0	0	1
<i>Spillovers (nsoe to soe)_i</i>	23,051	0.195	0.396	0	0	0
<i>Spillovers (soe to soe)_i</i>	23,051	0.179	0.383	0	0	0
<i>Legal_i</i>	23,051	0.060	0.237	0	0	0
<i>GDP_i</i>	23,051	0.068	0.252	0	0	0
<i>HHI_i</i>	23,051	0.092	0.289	0	0	0
<i>Size rival_i</i>	23,051	22.412	1.479	21.398	22.185	23.177
<i>Size event_i</i>	23,051	23.028	1.784	21.717	22.646	24.064
<i>Leverage_i</i>	23,051	0.426	0.219	0.249	0.411	0.582
<i>Top_i</i>	23,051	0.346	0.147	0.235	0.326	0.443
<i>Similar_i</i>	23,051	0.419	0.163	0.306	0.417	0.530

Note: This table presents the summary statistics of the variables used in the cross-sectional regression tests.

that equals 1 if an event firm and its industry rival are in regions with weak legal protection, and 0 otherwise. *GDP_i* equals 1 if an event firm and its industry rival are located in less developed regions, and 0 otherwise. As demonstrated in columns (4) and (5) of Table 10, the coefficients of both *Legal_i* and *GDP_i* are significant and negative. This suggests that rivals experience more substantial negative stock market reactions when they and event firms are in regions with weak legal protection or underdeveloped economies, supporting Hypothesis 4's

argument.

5.2.3. Impact of industry competition on contagion stock returns

This investigation continues into the effect of the degree of industry competition on scandal-related returns. Following Lang and Stulz (1992), we use the Herfindahl index¹⁷ as a proxy for the degree of industry competition. Industries are divided into high- and low-competition based on the median of the Herfindahl index. *HHI_i* is an indicator that equals 1 if a firm is in a less competitive industry, and 0 otherwise. The empirical results, reported in column (6) of Table 10, document that the coefficient of *HHI_i* is negative and statistically significant at the 1% level. Thus, rivals in low-competition industries experience a greater negative influence than those in high-competition industries, providing evidence to support our common risk hypothesis.

6. Industry Rivals' responses to environmental and social scandals

The prior analysis notes that ES scandals adversely affect industry competitors via risk channels. However, these effects can be mitigated by the ES activities of the rivals. This section extends the investigation to gauge whether the value attributed to ES encourages competitors with subpar ES performance to enhance their subsequent ES strategies to diminish the negative impact engendered by industry scandals. To address this query, we propose the following regression model.

¹⁷ The Herfindahl index, a traditional measure of concentration, is widely used in previous studies.

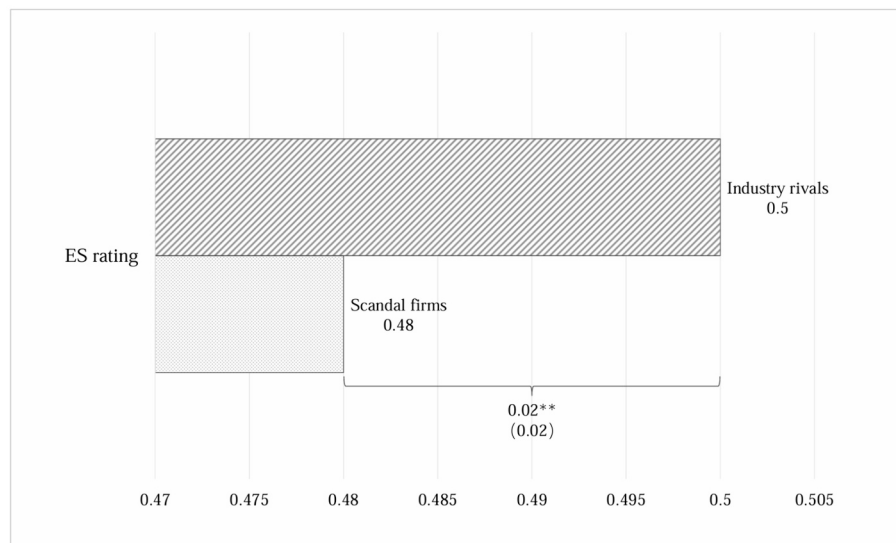


Fig. 3. Environmental and social (ES) performance by group.

Note: This figure plots corporate ES performance by group. The two groups comprise scandal firms and industry rivals without ES scandals from 2014 to 2021. *p*-values are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 9

Effect of rivals' environmental and social (ES) performance on contagion stock returns.

Dependent Variables	(1)	(2)
	CAR rival _{<i>i</i>}	CAR rival _{<i>i</i>}
ES rival _{<i>i</i>}	0.474*** (2.78)	0.299 (1.50)
ES rival _{<i>i</i>} × Disclosure rival _{<i>i</i>}		0.704*** (2.78)
Disclosure rival _{<i>i</i>}		−0.439*** (−2.84)
ES event _{<i>i</i>}	−0.022 (−0.14)	−0.022 (−0.14)
CAR event _{<i>i</i>}	0.083*** (9.73)	0.083*** (9.71)
Size event _{<i>i</i>}	0.025 (0.81)	0.025 (0.85)
Size rival _{<i>i</i>}	0.045 (1.25)	0.050 (1.27)
Top _{<i>i</i>}	−0.089 (−0.32)	−0.100 (−0.36)
Leverage _{<i>i</i>}	−0.290 (−1.12)	−0.287 (−1.11)
Similar _{<i>i</i>}	−0.658** (−1.98)	−0.648* (−1.95)
Industry-fixed effect	Yes	Yes
Year-fixed effect	Yes	Yes
# of observations	23,051	23,051
R ²	0.017	0.017

Note: This table presents the impact of rivals' ES performance on Cumulative Abnormal Returns (CARs). The dependent variable, CAR rival_{*i*}, represents the five-day cumulative abnormal return of individual rivals. ES rival_{*i*} denotes the ES ratings of rival firms from the previous year, sourced from the IIGF ESG databases. Disclosure rival_{*i*} is an indicator that equals 1 if rivals disclosed ESG reports in the previous year, and 0 otherwise. Control variables encompass the ES performance (ES event_{*i*}), five-day cumulative abnormal returns (CAR event_{*i*}), and firm size (Size event_{*i*}) of event firms, along with the largest shareholder's ownership (Top_{*i*}), leverage ratio (Leverage_{*i*}), and firm size (Size rival_{*i*}) of rival firms, and the similarity between industry rivals and event firms (Similar_{*i*}). Further details about the variables are provided in Table A2 in the Appendix. The regression estimates are derived from ordinary least squares, controlling for both industry- and year-fixed effects, and standard errors are clustered by firm across all models. The regression constant is omitted for brevity. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

$$\Delta ES_{i,t+1} = \alpha + \beta ES Low_{i,t} + \delta Y_{i,t} + Firm\ dummies + Year\ dummies + \varepsilon_{i,t} \quad (2)$$

where $\Delta ES_{i,t+1}$ represents the change in rivals' ES ratings in the subsequent year. A higher value indicates a greater improvement in the rivals' ES performance. The independent variable, $ES Low_{i,t}$, is an indicator that is set to 1 for rivals with ES ratings below the sample median in a given year, and 0 otherwise. The control variables, $Y_{i,t}$ include the rivals' leverage ratio, firm size, and the ownership of the largest shareholders. We also incorporate firm- and year-fixed effects to account for unobservable effects and cluster the standard errors by firm.

In column (1) of Table 11, the coefficient of $ES Low_{i,t}$ is 0.046 and statistically significant at the 1% level. This suggests that rivals with weaker ES performance increase their ES scores in the following year by an average of 0.046 relative to those with stronger performance. This increase is statistically significant and economically substantial, equating to nearly 10% of the average ES score of rival firms (0.451). Given that ESG rating agencies in China do not have a fiduciary relationship with the firms they rate, it is improbable that the increase in ES scores for underperforming rivals is due to pressure exerted on the rating agencies to make alterations. A plausible explanation for this result is that rivals with low ES ratings enhance their subsequent ES performance in response to other external shocks (e.g., government policies).

We continue to categorise our sample into two groups based on whether the rivals are adversely affected by ES scandals. Specifically, if the low ES rivals opt to improve their subsequent ES adoption to gain the risk-averse advantage of ES in industry scandals, we would anticipate a more pronounced positive relationship in rivals negatively affected by scandals than in those unaffected. In practice, we group the rivals in two distinct ways. Firstly, if a rival has a negative five-day CAR on average in a given year, it is considered negatively affected by industry ES scandals and is categorised in the negative CAR group; otherwise, it is in the non-negative CAR group. Secondly, if a rival has a negative five-day CAR in at least one industry scandal in a given year, it is placed in the negative CAR group; otherwise, it is in the non-negative CAR group.

As shown in columns (2) to (5) in Table 11, the results are consistent regardless of the grouping methodology used. We find that the coefficient of $ES Low_{i,t}$ is significant and positive for rivals in the negative CAR group but insignificant for those in the non-negative CAR group. These results align with our expectations. Following Cleary's method (Cleary, 1999), we employ the bootstrapping resampling method to test the

Table 10
Further analyses.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variables	CAR rival _i	CAR rival _i	CAR rival _i	CAR rival _i	CAR rival _i	CAR rival _i
SOE rival _i	0.252*** (3.01)					
Spillovers (nsoe to soe) _i		0.373*** (4.04)				
Spillovers (soe to soe) _i			−0.073 (−0.63)			
Legal _i				−0.529*** (−3.10)		
GDP _i					−0.827*** (−5.08)	
HHI _i						−0.357*** (−2.81)
CAR event _i	0.083*** (9.74)	0.084*** (9.83)	0.083*** (9.74)	0.083*** (9.78)	0.083*** (9.78)	0.079*** (9.80)
Size event _i	0.025 (0.83)	0.035 (1.18)	0.027 (0.89)	0.025 (0.83)	0.025 (0.83)	−0.005 (−0.21)
Size rival _i	0.047 (1.28)	0.048 (1.33)	0.066* (1.81)	0.061* (1.69)	0.060* (1.67)	0.005 (0.15)
Top _i	−0.195 (−0.70)	−0.170 (−0.61)	−0.073 (−0.26)	−0.094 (−0.34)	−0.105 (−0.38)	−0.358 (−1.31)
Leverage _i	−0.386 (−1.48)	−0.373 (−1.44)	−0.280 (−1.08)	−0.246 (−0.95)	−0.233 (−0.90)	−0.338 (−1.38)
Similar _i	−0.684** (−2.04)	−0.631* (−1.89)	−0.628* (−1.87)	−0.610* (−1.82)	−0.621* (−1.86)	−0.748** (−2.53)
Industry-fixed effect	Yes	Yes	Yes	Yes	Yes	No
Year-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
# of observations	23,051	23,051	23,051	23,051	23,051	23,051
R ²	0.017	0.017	0.017	0.017	0.018	0.007

Note: This table examines the influence of state ownership, external environment, and industry competition on Cumulative Abnormal Returns (CARs). The dependent variable, CAR rival_i, is the five-day cumulative abnormal return of individual rivals. SOE rival_i is an indicator that equals 1 if rivals are state-owned enterprises (SOEs), and 0 otherwise. Spillovers (nsoe to soe)_i and Spillovers (soe to soe)_i are indicators that equal 1 if event firms are non-SOEs and their rival firms are SOEs, and if both event firms and their rival firms are SOEs, respectively. Legal_i and GDP_i are indicators that equal 1 if both event firms and their industry rivals are located in regions with a weak legal environment, and in less economically developed regions, respectively. HHI_i is an indicator that equals 1 if firms operate in low-competition industries. The control variables include CAR event_i, Size event_i, Size rival_i, Top_i, Leverage_i, and Similar_i. Both industry- and year-fixed effects are controlled for, and standard errors are clustered by firm across all models. The regression estimates are derived from ordinary least squares. The regression constant is omitted for brevity. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 11
Subsequent environmental and social (ES) performance.

	(1)	(2)	(3)	(4)	(5)
	Full sample	CAR (Average)		CAR (Lowest)	
		Negative	Non-negative	Negative	Non-negative
Dependent variables	ΔES _{i,t+1}	ΔES _{i,t+1}	ΔES _{i,t+1}	ΔES _{i,t+1}	ΔES _{i,t+1}
ES Low _{i,t}	0.046*** (3.92)	0.049*** (2.77)	0.018 (0.83)	0.056*** (3.74)	0.031 (0.99)
Size rival _{i,t}	−0.013 (−1.49)	−0.009 (−0.79)	−0.025 (−1.06)	−0.014 (−1.30)	−0.027 (−0.77)
Top _{i,t}	−0.039 (−0.67)	−0.020 (−0.25)	−0.029 (−0.26)	−0.041 (−0.55)	0.078 (0.50)
Leverage _{i,t}	0.034 (1.09)	0.053 (1.16)	0.035 (0.47)	0.054 (1.28)	0.108 (1.03)
Firm-fixed effect	Yes	Yes	Yes	Yes	Yes
Year-fixed effect	Yes	Yes	Yes	Yes	Yes
# of observations	8910	4562	2721	6080	1301
R ²	0.185	0.289	0.334	0.230	0.362
Empirical <i>p</i> -value			0.042**		0.016**

Note: This table investigates the influence of ES scandals on industry rivals' subsequent ES performance. The dependent variable, ΔES_{i,t+1}, is the change in rivals' ES ratings in the following year. The independent variable, ES Low_{i,t}, is an indicator that equals 1 for rivals if the ES rating is below the sample median in a given year. Rival firms are divided into two groups based on whether they are negatively influenced by ES scandals. In columns (2) and (3), a firm is in the negative group if it has a negative CAR on average in a given year; otherwise, it is in the non-negative group. In columns (4) and (5), a firm is in the negative group if it has a negative CAR in at least one industry scandal in a given year; otherwise, it is in the non-negative group. The control variables include rivals' firm size (Size rival_{i,t}), leverage ratio (Leverage_{i,t}), and largest shareholder's ownership (Top_{i,t}). Both firm- and year-fixed effects are controlled for, and standard errors are clustered by firm across all models. Difference tests are conducted with the null hypothesis that the coefficients of ES Low_{i,t} are the same in columns (2) and (3) or in columns (4) and (5). Following Cleary (1999), a bootstrapping approach is used, and the empirical *p*-value is calculated with a 500-times bootstrapping procedure to estimate the likelihood of obtaining the null hypotheses. The regression constant is omitted for brevity. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

statistical difference between the coefficients of $ES\ Low_{i,t}$ in the two groups. The results indicate that the differences are statistically different from 0 at a 5% significance level. Therefore, we conclude that the positive impact of ES activities in industry scandals incentivises low ES rivals to take measures to improve subsequent ES ratings, thereby confirming a potential motivation for firms to enhance ES performance.

7. Conclusion

This study examines the industry effect of ES scandals among Chinese A-share listed firms. We observe that industry rivals incur negative ARs following the announcement of ES scandals, suggesting that investors perceive firms within the same industry as having homogeneous behaviour and sharing similar ES risks. These findings remain robust following a series of endogeneity and robustness tests. Further, we investigate the common risk channel underlying the contagion effect. Specifically, we examine the influence of rivals' ES performance on the negative market reaction induced by industry ES scandals. Empirical results indicate a significant positive correlation between rivals' ES ratings and their five-day CARs. This relationship is more pronounced among rivals that disclose ES information. We also find that state ownership, external environment, and industry competition can affect the spillover effect of ES scandals, supporting a risk-based explanation. Additionally, we explore whether the risk-averse advantage of ES in

scandals influences rivals' subsequent ES strategies. Evidence suggests that rivals with low ES significantly enhance their ES performance in the following year, motivated by the value of ES in industry scandals.

Our research enriches the understanding of the effects of ES scandals on industry rivals and underscores the positive role of ES in these spillover effects. Further, our findings have immediate implications for firm managers to view ES as a risk mitigator and to enhance social capital in response to adverse shocks. They also provide insights for investors and policymakers regarding the tangible impacts of ES on firm valuation, underscoring the importance for investors to integrate ES considerations into their investment strategies and for policymakers to advocate sustainable development.

Our study is subject to several limitations. First, we focus only on the equity market reactions of listed rival firms. Many private competitors also experience industry-related scandals. Exploring the responses of private competitors to ES scandals would be a valuable extension. Second, our analysis is confined to equity market responses. Investigating responses in other financial markets, such as bond markets and credit default swap markets, would offer additional valuable insights.

Data availability

The authors do not have permission to share data.

Appendix A. Appendix

Table A1

International Institute of Green Finance (IIGF) environmental, social, and governance (ESG) indicators.

Primary indicator	Secondary indicator	Qualitative/Quantitative
Environment	Energy-saving and emission reduction	Qualitative
	Pollution treatment	Qualitative
	Publicity of greening and environmental protection	Qualitative
	Green office	Qualitative
	Green design*	Qualitative
	Green technology*	Qualitative
	Green supply*	Qualitative
	Green production*	Qualitative
	Green financial products*	Qualitative
	Green revenue*	Quantitative
	Negative news	Quantitative
	Penalty	Quantitative
Social	Charity activities	Qualitative
	Community relations	Qualitative
	Employee relations	Qualitative
	Consumer relations	Qualitative
	Supplier relations	Qualitative
	Negative news	Quantitative
Governance	Organisation structure	Qualitative
	Investor relations	Qualitative
	Information transparency	Qualitative
	Technical innovation	Qualitative
	Risk management	Qualitative
	Negative news	Quantitative
	Penalty	Quantitative

Note: This table reports detailed information on ESG indicators in the IIGF ESG rating system.

* Secondary indicators representing industry-specific indicators.

Table A2
Variable explanation.

Variable	Definition
<i>CAR rival</i>	Industry rivals' cumulative abnormal returns in the event window (−2, 2), multiplied by 100.
<i>CAR event</i>	Event firms' cumulative abnormal returns in the event window (−2, 2), multiplied by 100.
<i>ES rival</i>	Industry rivals' ES ratings. Source: <i>IIGF ESG database</i> .
<i>ES event</i>	Event firms' ES ratings. Source: <i>IIGF ESG database</i> .
ΔES	The change of industry rivals' ES scores in the following year. Source: <i>IIGF ESG database</i> .
<i>Disclosure rival</i>	It is an indicator that equals 1 for rivals who disclose ESG reports, and 0 otherwise. Source: <i>IIGF ESG database</i> .
<i>SOE rival</i>	It is an indicator that equals 1 if a rival firm is an SOE, and zero otherwise.
<i>Spillovers (nonsoe to soe)</i>	It is an indicator that equals 1 if an event firm is a non-SOE and its industry rival is an SOE, and 0 otherwise.
<i>Spillovers (soe to soe)</i>	It is an indicator that equals 1 if an event firm and its industry rival are SOEs, and 0 otherwise.
<i>Legal</i>	It is an indicator that equals 1 if an event firm and its industry rival are in regions with weak legal environments, and 0 otherwise. Based on the median of the Legal Environment Index value, we divide the provinces into regions with weak and strong legal protection.
<i>GDP</i>	It is an indicator that equals 1 if an event firm and its industry rival are in less developed regions, and 0 otherwise. Based on the median GDP per capita, we divide the provinces into two groups: less and more developed regions.
<i>HHI</i>	It is an indicator that equals 1 if a firm is in a less competitive industry, and 0 otherwise. Based on the median of Herfindahl index, we divide the industries into high- and low-competition industries. $Herfindahl\ index = \sum (Sales_i / Sales)^2$
<i>Top</i>	Rival firms' largest shareholder's ownership at the end of the previous year.
<i>Leverage</i>	Rival firms' financial leverage ratio at the end of the previous year. Financial leverage ratio is the ratio of short-term debt plus long-term debt to total assets.
<i>Size rival</i>	It is the size of rival firms and is defined as the logarithm of total assets at the end of the previous year.
<i>Size event</i>	It is the size of event firms and is defined as the logarithm of total assets at the end of the previous year.
<i>Similar</i>	Following Lang and Stulz (1992) , <i>Similar</i> is measured as the correlation of stock returns between event firms and industry rivals. We calculate the correlation based on stock returns from 220 to 11 days before the ES scandals.

Note: This table details the main variables in the cross-sectional regressions.

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